vlsi world

Saturday, 4 February 2012

Verilog Codes for different COUNTERS

Verilog code for a 4-bit unsigned up counter with asynchronous clear.

```
module counter (clk, clr, q);
input        clk, clr;
output [3:0] q;
reg    [3:0] tmp;
always @(posedge clk or posedge clr)
begin
   if (clr)
      tmp <= 4'b0000;
   else
      tmp <= tmp + 1'b1;
end
   assign q = tmp;
endmodule
```

Verilog code for a 4-bit unsigned down counter with synchronous set.

```
module counter (clk, s, q);
input        clk, s;
output [3:0] q;
reg    [3:0] tmp;
always @(posedge clk)
begin
   if (s)
      tmp <= 4'b1111;
   else
      tmp <= tmp - 1'b1;
end
   assign q = tmp;
endmodule
```

Verilog code for a 4-bit unsigned up counter with an asynchronous load from the primary input.

```
module counter (clk, load, d, q);
input        clk, load;
input  [3:0] d;
output [3:0] q;
reg    [3:0] tmp;
always @(posedge clk or posedge load)
begin
   if (load)
      tmp <= d;
   else
      tmp <= tmp + 1'b1;
end
   assign q = tmp;
endmodule
```

Verilog code for a 4-bit unsigned up counter with a synchronous load with a constant.

```
module counter (clk, sload, q);
input        clk, sload;
output [3:0] q;
reg    [3:0] tmp;
always @(posedge clk)
begin
   if (sload)
            tmp <= 4'b1010;
   else
      tmp <= tmp + 1'b1;
end
   assign q = tmp;
```

Blog Archive

▼ 2012 (35)
▼ February (28)
vlsiupdatez.blogspot.com
VLSI SEMINAR TOPICS
Need of Textbooks softcopies
Digital questions -6
Digital questions-5
Digital questions - 4
Digital questions -3
Digital questions-2
Digital questions -1
Electronics Hardware Questions
Verilog Code for FSMs
Verilog Code for RAM & ROM
Verilog code for comparator
Verilog code for adders/subtractors
Verilog code for different MUX
Verilog codes for different Shift-registers
verilog code for ACCUMULATOR
Verilog Codes for different COUNTERS
Verilog code for a tristate element
Verilog Code for different LATCHES
Verilog code for a 4-bit register
verilog code for different FLIP-FLOPS
Verilog code for encoder using ASSIGN
Verilog Code for PRIORITY ENCODER using IF-ELSE
verilog code for ENCODER using CASE
verilog code for ENCODER using IF-ELSE
verilog code for DECODER using Assign statement
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► January (7)

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About Me

kumar

I am kumar from Andhra pradesh, INDIA. I have completed M.Tech in VLSI. Getting M.tech degree in VLSI is very easy because now a days all colleges are offering this course. But, entering into VLSI industry is not that much easy as compared to software. This is because of colleges are not having proper infrastructure or proper tools as required for VLSI course. So they are Lag in campus placements too. Only IITs, NIITs, and some other universities are good in course and jobs. Even I have faced lot of problems to enter into VLSI industry. So this blog contains all the information exclusively for freshers, latest technologies in VLSI, latest updates,Openings for freshers, interview questions for freshers, queries.

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```
endmodule
```

Verilog code for a 4-bit unsigned up counter with an asynchronous clear and a clock enable.

```
module counter (clk, clr, ce, q);
input        clk, clr, ce;
output [3:0] q;
reg    [3:0] tmp;
always @(posedge clk or posedge clr)
begin
   if (clr)
      tmp <= 4'b0000;
   else if (ce)
      tmp <= tmp + 1'b1;
end
   assign q = tmp;
       endmodule
```

Verilog code for a 4-bit unsigned up/down counter with an asynchronous clear.

```
module counter (clk, clr, up_down, q);
input        clk, clr, up_down;
output [3:0] q;
reg    [3:0] tmp;
always @(posedge clk or posedge clr)
begin
   if (clr)
      tmp <= 4'b0000;
   else if (up_down)
      tmp <= tmp + 1'b1;
   else
      tmp <= tmp - 1'b1;
end
   assign q = tmp;
       endmodule
```

e Verilog code for a 4-bit signed up counter with an asynchronous reset.

```
module counter (clk, clr, q);
input              clk, clr;
output signed [3:0] q;
reg    signed [3:0] tmp;
always @ (posedge clk or posedge clr)
begin
   if (clr)
      tmp <= 4'b0000;
   else
      tmp <= tmp + 1'b1;
end
   assign q = tmp;
endmodule
```

Verilog code for a 4-bit signed up counter with an asynchronous reset and a modulo maximum.

```
module counter (clk, clr, q);
parameter MAX_SQRT = 4, MAX = (MAX_SQRT*MAX_SQRT);
input                clk, clr;
output [MAX_SQRT-1:0] q;
reg    [MAX_SQRT-1:0] cnt;
always @ (posedge clk or posedge clr)
begin
   if (clr)
      cnt <= 0;
   else
      cnt <= (cnt + 1) %MAX;
end
   assign q = cnt;
endmodule
```

Posted by kumar at 01:23

3 comments:



HIMANSHU SHEKHAR 7 November 2013 08:40

hello sir,
i want code of 3 bit up/down synchronous counter in verilog..

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will u pls help me
if not give me some idea how to write its code in verilog..because i had design the ckt but i don't know how to write code for this ckt.

Reply

Unknown 9 June 2014 04:22

I second Himanshu's request. Along with a gated D-latch example.

Reply

Ishita Biswas 24 June 2014 23:11

hello sir,
I am doing processor modelling project. Will you please give me the codes of memory unit in verilog where there is no user input.
Thank you

Reply

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